

# A METAMODEL FOR ONTOLOGICALLY HETEROGENEOUS SOCIAL DYNAMICS

## PART I: THE HIGHEST ABSTRACT LAYER

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### Abstract

This is the first paper of a series. It fixes the highest abstract layer, and it goes no further.

Dynamical modeling of social systems ordinarily begins by fixing a state space. This is already a commitment, and across domains it is one that cannot be met. The states of a normative order, a power configuration, a value system, and a network of recognition inhabit no common manifold and admit no common metric—and the obstacle is not that such a space is hard to construct but that these are not objects of a kind that admit a common placement. The heterogeneity is ontological, not practical, and it is not answered by better instruments. The proposal here is to begin one layer lower, at a level where change, structure, constraint, and difference are defined but not yet realized. Four axioms are given. The carrier of states is left without structure. The mechanisms acting upon it form a semigroup: associative, without inverses, without an identity. A congruence records what any mechanism is obliged to preserve. And difference is defined—not as a residue, not as a distance—as the set of distinctions a mechanism collapses. From these, four propositions follow that were not assumed: collapse is irreversible; a group of mechanisms has no collapse at all, and hence no history; a mechanism is non-destructive exactly when it is injective modulo the congruence; and collapse accumulates monotonically under composition. The absence of inverses is shown to be the load-bearing commitment. This paper does not instantiate the metamodel, does not apply it, and derives no substantive result about any particular social system. Those belong to subsequent parts. What is established here is only what a social dynamical system must be able to say, and the price of saying it.

**Keywords:** dynamical systems; social dynamics; semigroup action; congruence; collapse; irreversibility; formal social theory; generative relational being.

**AI usage statement:** The ideas, claims, and theoretical framework of this paper originate with the author. AI tools were used as an aid in drafting, in the location and verification of sources, and in revision. The origin of each idea has been verified by the author, who takes full responsibility for the content.

## RECAP

The metamodel in full.  $\mathcal{S}$  is a set without further structure;  $\mathcal{M}$  is a semigroup, with neither identity nor inverses;  $\sim$  is an equivalence relation on  $\mathcal{S}$ .

|              |                                                                                                      |
|--------------|------------------------------------------------------------------------------------------------------|
| (Change)     | $\alpha : \mathcal{M} \times \mathcal{S} \longrightarrow \mathcal{S}, \quad (m, x) \longmapsto m(x)$ |
| (Structure)  | $\alpha(m_1 m_2, x) = \alpha(m_1, \alpha(m_2, x))$                                                   |
| (Constraint) | $x \sim y \implies \alpha(m, x) \sim \alpha(m, y)$                                                   |
| (Difference) | $\Delta(m) = \{ (x, y) : x \not\sim y \wedge \alpha(m, x) \sim \alpha(m, y) \}$                      |

Sections 1–3 give the reasons for these four lines, §4 their consequences, and §5 what they decline to settle. The consequences, which were not assumed, are that collapse is irreversible, that irreversibility requires the absence of inverses, and that a mechanism destroys nothing when it is injective modulo  $\sim$ .

## 1 INTRODUCTION

### 1.1 THE FORMALISM IS LAYERED; THIS PAPER GIVES THE TOPMOST LAYER

The programme to which this paper belongs proposes a formalism for social dynamics at several levels of abstraction. At the highest, one says what any model must be able to express, without saying what it expresses. Below it, one says what structures a model may draw upon—an order, a metric, an algebra—and at what cost. Below that, one exhibits models. The levels are not competing accounts of the same thing. They stand in a relation of dependence: what is settled above constrains what may be built beneath, and what is built beneath must answer to it.

This is the first paper of the series, and it gives only the highest layer. It does not instantiate the framework, does not apply it to a domain, and derives no substantive result about any particular social system. Those belong to the parts that follow.

The reason for beginning at the top, and for treating it separately, is that the commitments made at this layer are the ones that later become invisible. A model which ranks has assumed comparability; a model which measures has assumed symmetry; a model which aggregates has assumed an arithmetic. Each of these is a decision, and each, once buried inside a concrete formalism, is difficult to retrieve and impossible to contest. Isolating the highest layer is a way of making the decisions available for inspection before they are made.

### 1.2 WRITING $x$ ALREADY STIPULATES WHAT COUNTS AS A DIFFERENCE

A dynamical system, in its received form, is a triple: a state space, a rule of evolution, and a parameter. The rule is written

$$\dot{x} = f(x; \theta),$$

and the reading it invites is that the left side records change and the right side records the structure from which change proceeds. That reading is correct as far as it goes. What it conceals is the cost of the first term. To write  $x$  at all is to have fixed a space in which  $x$  lives, and with

it a stipulation of what counts as a difference between one state and another: whether such differences can be measured, whether they can be compared, whether they vary continuously.

For the systems at issue here, that stipulation cannot be met. The state of a normative order is a configuration of rules and expectations; the state of a power system is a distribution of asymmetries; the state of a value system is an ordering of what matters; the state of a recognition network is a pattern of who is seen. These do not sit in a common space, and they admit no common metric.

The point requires care, because it is stronger than it may appear. The difficulty is not that a common space is hard to construct, which would be a difficulty of instruments and would be answered by better ones. It is that a rule and a dependence are not two values of one variable, and no refinement of measurement will make them so. The heterogeneity of social states is ontological rather than practical, and a heterogeneity of that kind is not overcome; it is either respected or overwritten. It is the claim that these states *are* commensurable—that some scale  $m : X \rightarrow \mathbb{R}$  exists—which the theory of value pluralism has spent a century contesting, and a modeling strategy that begins by assuming the space begins by assuming away what is contested.

### 1.3 CHANGE, STRUCTURE, CONSTRAINT, AND DIFFERENCE NEED NO STATE SPACE

The response proposed is not to find a cleverer space. It is to descend below the point at which a space is required.

Four things must be sayable of any social dynamical system. It must be able to say how it changes. It must be able to say how its changes compose. It must be able to say what its changes are constrained to preserve. And it must be able to say what its changes destroy. These are, respectively, *change*, *structure*, *constraint*, and *difference*.

The claim of this paper is not that these four are novel. It is that they are, taken together, sufficient, and that they can be given without assuming a state space, a metric, an order, an arithmetic, or a zero. This matters, because each of those assumptions, once made, quietly decides a question the theory was meant to leave open. An order on states presumes commensurability. A metric presumes symmetry, and therefore cannot distinguish betraying from being betrayed. A zero presumes that what a transformation leaves behind is a quantity capable of being null, and so decides, in advance of any argument, that loss is a magnitude.

The metamodel given in §3 makes none of these choices. In exchange, it yields results that were not put in by hand: that collapse is irreversible, that irreversibility is possible only because mechanisms lack inverses, and that a mechanism destroys nothing exactly when it is injective modulo the congruence.

### 1.4 LOSS PRECEDES GENERATIVITY IN THE ORDER OF DEPENDENCE

Instantiation is deferred. So is application. So, in particular, is the formal treatment of generativity, which is the term the wider programme exists to serve and which the four axioms of this paper do not provide. What they provide is its precondition: an account of what a mechanism can destroy. A framework that cannot say what is lost cannot say what it would mean to generate. The order of exposition follows the order of dependence, and loss comes first.

## 2 THE STATE SPACE CANNOT COME FIRST

### 2.1 SOCIAL STATES ARE HETEROGENEOUS IN KIND, NOT IN SCALE

Consider four systems and the state each would require.

| SYSTEM              | STATE                                             |
|---------------------|---------------------------------------------------|
| normative order     | configuration of rules, obligations, expectations |
| power system        | distribution of dependence and authority          |
| value system        | ordering of what is held to matter                |
| recognition network | pattern of who is acknowledged as a subject       |

No embedding of these into a common vector space is available, and the ones that have been proposed are precisely the objects of critique: utility, price, and the composite index each purchase comparability at the cost of the differences they were meant to preserve.

It is worth being exact about the sense in which this heterogeneity is *ontological* and not merely practical, since the two are easily confused and only the former has consequences for the form of a theory. Practical heterogeneity would mean that a common space exists but is hard to construct—that the data are poor, the scales incommensurate in practice, the measurement instruments crude. Such a difficulty is answered by better instruments, and it licenses the modeler to proceed as though the space were there, since in principle it is.

The heterogeneity at issue is not of that kind. It is not that a common space is hard to find but that the objects to be placed in it are not of a kind that admits a common placement. A rule and a dependence are not two values of one variable measured badly; they are not two values of one variable. A pattern of recognition is not a coarse rendering of a distribution of authority. These are distinct modes of being, and the question of how far one lies from another is not a hard question but a malformed one. It is in this sense that the demand for a common state space is not a technical inconvenience but a substantive commitment—indeed an ontological one, asserted in the guise of a modeling convention—and it should not be made by default.

### 2.2 ORDER PRESUMES COMMENSURABILITY, METRIC PRESUMES SYMMETRY, ZERO PRESUMES MAGNITUDE

**Order.** A partial order on  $\mathcal{S}$  furnishes suprema, and a supremum of two value states is a common measure of them. To assume a lattice is to have assumed commensurability.

**Metric.** A metric  $d$  on  $\mathcal{S}$  satisfies  $d(x, y) = d(y, x)$ . Symmetry cannot represent a directed act. Under a metric, a mechanism and its reception are the same distance apart, and the asymmetry of the relational act has been erased before any question about it can be posed.

**Zero.** An additive structure with a zero element makes the residue of a transformation a quantity that may or may not vanish. This is a determinate philosophical position—the position on which what is lost is a magnitude, and so, in principle, restorable. The rival position holds that what is lost has no place in the new structure at all. The metamodel must be able to host both, and therefore may assume neither.

### 2.3 FIXING THE MECHANISMS LEAVES THE CARRIER FREE

The path taken here inverts the usual order of construction. Rather than fixing states and deriving the transformations that act upon them, we fix the transformations and leave the states

unstructured. This is not new; it is the one by which the algebraic approach to dynamics proceeds. Its warrant is that the structure a theory requires can be carried by the acting objects, leaving the carrier as a set on which they act.

### 3 CHANGE, STRUCTURE, CONSTRAINT, AND DIFFERENCE ARE DEFINABLE WITHOUT A STRUCTURED CARRIER

#### 3.1 THE CARRIER IS A BARE SET, THE MECHANISMS CARRY THE STRUCTURE

**Definition 1 (Primitives).** A *social dynamical metamodel* is a triple  $(S, \mathcal{M}, \sim)$  where

- $S$  is a set, the *carrier*. It is assumed to have no further structure: no order, no metric, no algebraic operation, no distinguished element.
- $\mathcal{M}$  is a set, the *mechanisms*.
- $\sim$  is an equivalence relation on  $S$ .

Elements of  $S$  are states, in the weakest sense the word can bear: whatever a given domain takes a configuration to be. Elements of  $\mathcal{M}$  are mechanisms: whatever, in that domain, acts upon a configuration and yields another. Nothing is presumed about what either is made of.

#### 3.2 THE MECHANISMS FORM A SEMIGROUP, AND $\Delta$ IS DERIVED FROM $\alpha$ AND $\sim$

**Axiom 1 (Change).** There is a map

$$\alpha : \mathcal{M} \times S \longrightarrow S, \quad (m, x) \longmapsto m(x).$$

A mechanism acts on a state and returns a state. This is all that change is required to be.

**Axiom 2 (Structure).**  $\mathcal{M}$  carries an associative composition

$$\mathcal{M} \times \mathcal{M} \rightarrow \mathcal{M}, \quad (m_1, m_2) \mapsto m_1 m_2, \quad (m_1 m_2) m_3 = m_1 (m_2 m_3),$$

and  $\alpha$  is an action of it:

$$\alpha(m_1 m_2, x) = \alpha(m_1, \alpha(m_2, x)).$$

$\mathcal{M}$  is a semigroup. It is *not* assumed to have an identity, and it is *not* assumed to have inverses.

Associativity is free: it says only that doing  $m_1$  then  $m_2$ , and then  $m_3$ , is the same as doing  $m_1$ , and then  $m_2$  followed by  $m_3$ . The two omissions are not free, and they are the substantive content of the axiom. Their consequences are drawn in §4.

**Axiom 3 (Constraint).**  $\sim$  is a congruence for the action:

$$x \sim y \implies \alpha(m, x) \sim \alpha(m, y) \quad \text{for all } m \in \mathcal{M}.$$

Read: whatever the system preserves, no mechanism may destroy. The axiom does not say what  $\sim$  is. It says that a mechanism which fails to respect it is not a mechanism of this system

at all.  $\sim$  is the system's law of admissibility, and its content is supplied by the domain.

**Axiom 4 (Difference).** For each  $m \in \mathcal{M}$ , the *collapse* of  $m$  is

$$\Delta(m) = \{ (x, y) \in \mathcal{S} \times \mathcal{S} : x \not\sim y \wedge \alpha(m, x) \sim \alpha(m, y) \}.$$

Read:  $\Delta(m)$  is the set of distinctions that  $m$  destroys. Two states that the system could tell apart become, after  $m$  has acted, states it cannot tell apart.

### 3.3 DIFFERENCE IS A LOST DISCRIMINATION, NOT A LEFTOVER ELEMENT

Three familiar formalizations of difference are rejected, each because it requires structure on  $\mathcal{S}$  that the metamodel declines to assume.

**Not a residuum.** The residuated-lattice operator,  $x \setminus m(x) = \max\{c : x \cdot c \leq m(x)\}$ , requires that  $\mathcal{S}$  carry both a multiplication and an order. It therefore requires that states be comparable, which is what the ontological heterogeneity of social states denies.

**Not a distance.** The metric-induced form,  $d(x, m(x))$ , requires a metric, and a metric is symmetric. It cannot represent a directed act, and the acts at issue are directed.

**Not a kernel.** The form  $\ker \pi = \{x : \pi(x) = 0\}$  requires a zero, hence an algebraic structure on  $\mathcal{S}$ . In a semigroup the kernel of a morphism is not a subset at all: it is a congruence. Axiom 4 takes that fact as its cue and defines difference as a relation rather than an element.

Difference, at this layer, is not a thing that is left over. It is a discrimination that has been lost.

The three rejections share a form. Each of the discarded operators returns an *element*, and in doing so commits the metamodel to a determinate view of what a remainder is—a view on which the traditions that have thought longest about remainders do not agree. What is at stake in this is taken up in §5.

$\Delta$  is a derived object. It is not a fifth primitive: it is defined entirely from  $\alpha$  and  $\sim$ , which is to say from Axioms 1 and 3.

## 4 IRREVERSIBILITY FOLLOWS FROM ASSOCIATIVITY AND THE CONGRUENCE ALONE

The axioms are weak. They nevertheless entail results that were not stipulated. This section gives them.

### 4.1 COLLAPSE IS IRREVERSIBLE

**Proposition 1 (Irreversibility of Collapse).** Let  $(x, y) \in \Delta(m)$ . Then for every  $m' \in \mathcal{M}$ ,

$$\alpha(m'm, x) \sim \alpha(m'm, y).$$

That is: no mechanism applied after  $m$  can restore the distinction  $m$  destroyed.

*Proof.* By definition of  $\Delta$ ,  $\alpha(m, x) \sim \alpha(m, y)$ . By Axiom 3, applying  $m'$  to congruent states yields congruent states, so  $\alpha(m', \alpha(m, x)) \sim \alpha(m', \alpha(m, y))$ . By Axiom 2, the left side is  $\alpha(m'm, x)$  and the right side is  $\alpha(m'm, y)$ .  $\square$

The proof uses only Axioms 2 and 3. Irreversibility is therefore not an extra postulate about the social world, appended because we know from experience that some things cannot be undone. It is what the associativity of mechanisms and the preservation of the congruence, taken together, already say.

## 4.2 INVERSES ANNIHILATE DIFFERENCE

**Proposition 2 (Groups have no history).** Suppose  $\mathcal{M}$  is a group acting on  $\mathcal{S}$  and  $\sim$  is a congruence. Then  $\Delta(m) = \emptyset$  for every  $m \in \mathcal{M}$ .

*Proof.* Suppose  $(x, y) \in \Delta(m)$ , so  $\alpha(m, x) \sim \alpha(m, y)$  and  $x \not\sim y$ . Applying  $m^{-1}$  and using Axiom 3,  $\alpha(m^{-1}m, x) \sim \alpha(m^{-1}m, y)$ , that is  $x \sim y$ , a contradiction.  $\square$

The absence of inverses is not a technical weakening. It is the condition under which a system can have a history at all. Where every mechanism can be undone, nothing is destroyed, nothing is preserved against destruction, and the question of what a transformation has cost does not arise. A group of mechanisms is a world without loss.

The two propositions stand in a definite relation. Proposition 1 says that what is collapsed stays collapsed. Proposition 2 says that this is possible only because mechanisms are not invertible. Taken together they locate the exact structural commitment on which everything else in this framework depends: Axiom 2's refusal of inverses.

## 4.3 A MECHANISM DESTROYS NOTHING IFF IT IS INJECTIVE MODULO $\sim$

**Proposition 3.**  $\Delta(m) = \emptyset$  if and only if the induced map  $\bar{m} : \mathcal{S}/\sim \rightarrow \mathcal{S}/\sim$  is injective.

*Proof.* Axiom 3 guarantees that  $\bar{m}$  is well defined on the quotient.  $\bar{m}$  fails to be injective exactly when there are classes  $[x] \neq [y]$  with  $\bar{m}[x] = \bar{m}[y]$ , that is, when there are states  $x \not\sim y$  with  $\alpha(m, x) \sim \alpha(m, y)$ , that is, when  $\Delta(m) \neq \emptyset$ .  $\square$

The proposition supplies a criterion. A mechanism destroys nothing if and only if it is injective modulo the congruence, so it is not necessary to inspect a mechanism's effects case by case in order to know whether it is destructive. It suffices to ask whether it separates what the system could already separate.

## 4.4 COMPOSITION ACCUMULATES

**Proposition 4 (Monotonicity of Collapse).** For all  $m, m' \in \mathcal{M}$ ,

$$\Delta(m) \subseteq \Delta(m'm).$$

*Proof.* Let  $(x, y) \in \Delta(m)$ . Then  $x \not\sim y$ , and by Proposition 1,  $\alpha(m'm, x) \sim \alpha(m'm, y)$ . Hence  $(x, y) \in \Delta(m'm)$ .  $\square$

Collapse, once incurred, is carried forward by every subsequent mechanism. The history of a system is thus a monotone accumulation of lost discriminations, and no mechanism in the semigroup can reduce it. This is the formal sense in which a social system does not return to a prior condition.

## 5 LOSS IS A MAGNITUDE ONLY IF THE CARRIER HAS AN ARITHMETIC

Axiom 4 defines difference as a relation. The natural alternative would define it as an element: a residue, a remainder, a something left over when a mechanism has acted. The alternative was declined, and the reason is that the two traditions with the strongest claim on the notion of a remainder—political economy and psychoanalysis—disagree about what a remainder is. An operator that returns an element settles their disagreement, and settles it by stipulation. What follows sets out the disagreement, shows where the notation would decide it, and shows what a relation-valued  $\Delta$  leaves open instead.

### 5.1 SURPLUS VALUE IS RESTORABLE; THE *OBJET A* HAS NO PLACE TO BE RESTORED TO

**The economic residue.** In the analysis of value, what a mechanism leaves over is a *magnitude*. Surplus value is the difference between the value a labourer produces and the value returned to him [12]; it is a quantity, it can in principle be computed, and—this is the point on which the whole normative force of the analysis rests—it could have been returned. That it was not is the injustice. The residue is thus an element of the same order as what was given: it has a location in the value system, it is expressible in the system's terms, and its restitution is intelligible even where it is not forthcoming. A theory of exploitation requires a residue of this kind, for otherwise it could not say what was taken.

**The psychoanalytic residue.** In Lacan the remainder is of a different kind altogether [14, 15]. The *objet petit a* is not a quantity of satisfaction withheld; it is what falls out of symbolization itself, the residue produced by the subject's entry into the symbolic order and precisely not representable within it. It is not that the object is scarce, or that it has been taken and might be given back. It has *no place* in the order that produced it. Its whole theoretical function depends on this: were it locatable, were it a magnitude, it could in principle be supplied, and the structure of desire that it sustains would collapse [16].

The economic residue is a magnitude that was not returned. The psychoanalytic residue is not a magnitude at all. The first can be restored. The second cannot, and the obstacle is not that restoration is difficult but that there is nothing of the right kind to restore.

### 5.2 AN ELEMENT-VALUED OPERATOR DECIDES BETWEEN THEM BY NOTATION

Suppose difference were defined, as it commonly is, by an operator returning an element:

$$\Delta(m, x) = x \setminus m(x) \quad \text{or} \quad \Delta(m, x) = d(x, m(x)) \quad \text{or} \quad \Delta(m) = \ker \pi_m = \{x : \pi_m(x) = 0\}.$$

Each of these requires structure on  $\mathcal{S}$ : the first an order and a multiplication, the second a metric, the third a zero. And each, in requiring it, has already chosen. An operator that returns an element of  $\mathcal{S}$ , or of some  $\mathcal{D}$  possessing a zero, has said that what is lost is a *something*—a magnitude that may be null, that has a place, that could conceivably be supplied. It has

decided for the economic residue and against the psychoanalytic one, and it has done so not by argument but by notation.

This is the kind of commitment the highest layer exists to hold open. Whether the remainder of a social mechanism is a restorable magnitude or an unrepresentable void is a substantive question, and one on which the traditions divide. A metamodel is arguably not the place to answer it, and one that answers it in its choice of symbols has answered it without argument.

### 5.3 A RELATION-VALUED $\Delta$ LEAVES BOTH READINGS OPEN

$$\Delta(m) = \{ (x, y) : x \not\sim y \wedge \alpha(m, x) \sim \alpha(m, y) \}$$

$\Delta$  returns a relation, not an element. It says which discriminations a mechanism destroyed. It does not say what, if anything, was left behind, and it requires no order, no metric, no arithmetic, and no zero in order to say what it says.

The two traditions are thereby left open, and each is recovered at instantiation rather than assumed at the outset:

- Where an instantiation supplies  $S$  with an additive structure, a magnitude can be associated with each collapsed pair, and the residue becomes a quantity—the economic reading. Restitution is then well posed, and one may ask whether it has been made.
- Where an instantiation supplies no such structure,  $\Delta(m)$  remains a set of pairs and nothing more. Two states have become indiscernible, and nothing of the right type exists to be returned—the psychoanalytic reading. Restitution is not withheld here; it is not *formulable*.

The metamodel is not neutral because it is empty. It is neutral because it locates the disagreement: the question of whether loss is a magnitude is the question of whether the carrier has an arithmetic. That is a decision about a particular social order, to be made and defended in the modeling of that order. It is not, so far as this paper can see, a decision about the form of social dynamics as such, and it should not be made by notation alone.

### 5.4 THE TWO READINGS LICENSE DIFFERENT DEMANDS

It should be said, though it exceeds the scope of this part, that the choice is consequential. A theory of justice built on the economic residue can demand restitution: something was taken, it is of a determinate size, and it can be given back. A theory built on the psychoanalytic residue cannot make this demand, and must ask instead for a different order—one whose congruence separates what the present one has conflated.

The metamodel does not adjudicate between them. It does establish that they are different demands, that they follow from different structural assumptions about the carrier, and that a theory which has not said which assumption it makes has not yet said what it is asking for.

## 6 INSTANTIATION SUPPLIES THE STRUCTURE THE AXIOMS WITHHOLD

The metamodel is deliberately too weak to compute with. It has no order, so it cannot rank. It has no metric, so it cannot measure. It has no arithmetic, so it cannot aggregate. Each of these is supplied only at instantiation, and each supply is a commitment. The purpose of separating the layers is to make the commitments visible, so that of any concrete model it may be asked: at which line did it assume comparability, and with what warrant.

## 6.1 EVERY INSTANTIATION IS A SEMIFLOW, AND ITS CONGRUENCE CARRIES THE NORMATIVE WEIGHT

To instantiate is to supply four things: a carrier  $S$ , a semigroup  $\mathcal{M}$ , an action  $\alpha$ , and a congruence  $\sim$ . The axioms then constrain what the instantiation may do, and the propositions of §4 hold in it automatically, without reproof.

Two constraints on instantiation follow directly from §4 and are worth stating here, since they hold for every instantiation and therefore belong to the highest layer rather than to any particular one.

**Semiflows, not flows.** A continuous-time flow with  $t \in \mathbb{R}$  is a group action, and by Proposition 2 it has  $\Delta \equiv \emptyset$ . Reversible dynamics has no collapse. This is the classical fact that a Hamiltonian system loses no information, and it is here a corollary rather than an observation. Any instantiation that is to model a system with a history must therefore be a semiflow,  $t \geq 0$ . The restriction is not a modeling convenience; it is forced.

**The congruence carries the normative weight.** In any instantiation,  $\sim$  is not a technical choice. It is the system's statement of what it is obliged to preserve, and it is the sole point at which normative content enters the formalism. Two configurations that a theory of justice would not have conflated are configurations that  $\sim$  is required to separate. Choosing  $\sim$  is therefore a substantive theoretical act, and one to be defended as such. The metamodel supplies the *form* of the obligation and none of its content.

## 6.2 INSTANTIATION, APPLICATION, AND GENERATIVITY ARE DEFERRED

Three things are outside the scope of Part I, and are named so that their absence is not mistaken for an oversight.

**No instantiation.** No carrier, semigroup, action, or congruence is exhibited for any concrete domain. The metamodel is presented as a set of conditions on models, not as a model.

**No application.** The framework is not read against any substantive problem in social or political theory. Where such readings are possible—and the author holds that they are, and that they alter the description of the phenomena—they require a determinate  $\sim$ , and hence an instantiation, and hence a later part.

**No account of generativity.** The axioms describe how mechanisms act on a fixed carrier. Nothing in them produces a state that was not already in  $S$ , nor a distinction that  $\sim$  did not already admit. Collapse has been given a formal home; emergence has not. This is a genuine limitation and is returned to in §6.

Part I establishes what a social dynamical system must be able to say. It does not say it.

## 7 THE AXIOMS CANNOT JUDGE A COLLAPSE, GENERATE THE NEW, OR EVOLVE THEIR OWN CONGRUENCE

The framework should be judged by what it cannot do, and four things it cannot do are named here.

**It cannot say that a collapse was wrong.**  $\Delta(m)$  records which distinctions a mechanism destroyed. It does not say they ought to have been kept. Some collapses are innocuous, and some are necessary: a system that could distinguish every state from every other would be a system incapable of generalization. The normative weight falls entirely on  $\sim$ , that is, on the prior determination of what a system is obliged to preserve. The metamodel supplies the form of that determination and nothing of its content.

**It cannot generate the new.** The axioms describe how mechanisms act on a fixed carrier. Nothing in them produces a state that was not already in  $\mathcal{S}$ , and nothing produces a distinction that  $\sim$  did not already admit. This is a genuine limitation for a programme whose central term is generativity. Collapse has been given a formal home; emergence has not. A candidate exists—the passage of a system into an inequivalent representation of the same algebra, where the new is not a rearrangement of the old but an accession to a sector formerly unreachable—but it requires more structure than four axioms provide, and it is not developed here.

**The congruence is exogenous.** Axiom 3 posits  $\sim$  and does not derive it. Yet in any actual system, what counts as a preserved distinction is itself an outcome of the system's history. A sharper framework would let  $\sim$  evolve, so that a mechanism could alter not only the states but the system's very capacity to discriminate among them. The immediate obstacle is that  $\Delta$  is defined in terms of  $\sim$ , and a moving  $\sim$  makes the definition circular. Resolving this is, in the author's view, the most important open problem in the framework, and it is the problem to which the next part is addressed.

**It is not, by itself, a theory.** A metamodel of this kind constrains the class of admissible models; it does not select among them, and it makes no prediction. Its use is diagnostic. It permits one to ask of any social model: what does it take a state to be, what does it take a mechanism to be, what does it oblige a mechanism to preserve, and what, therefore, does it permit a mechanism to destroy. A model that cannot answer the fourth question has not thereby avoided destruction. It has only declined to record it.

## 8 CONCLUSION

Four axioms have been given. A carrier without structure; a semigroup of mechanisms acting upon it, associative and without inverses; a congruence recording what the system is obliged to preserve; and a difference operator defined as the set of distinctions a mechanism collapses. From these, four propositions follow: that collapse is irreversible; that it is possible only in the absence of inverses; that a mechanism is non-destructive exactly when it is injective modulo the congruence; and that collapse accumulates monotonically under composition.

The propositions are not deep. They are three lines each. What is worth noticing is that they were not put in. The framework was constructed to be as weak as it could be made—no order, no metric, no arithmetic, no zero—and by assuming so little, it was able to obtain irreversibility as a theorem rather than posit it.

A system has a history because its mechanisms cannot be undone. It can lose what it was obliged to keep because its mechanisms can collapse distinctions the congruence required it to hold apart. These are the same fact, stated once algebraically and once normatively, and it is the burden of a formal social theory to show that they are the same.

That burden is not discharged here. This paper has established only the highest layer: what a social dynamical system must be able to say, and what it costs to say it. It has exhibited no instantiation, made no application, and given no account of the term—generativity—for the sake of which the wider programme exists.

The order of exposition, however, follows the order of dependence. A framework that cannot say what a mechanism destroys cannot say what it would mean to generate, since generation is intelligible only against the possibility of loss. Loss has now been given a form. It is a relation, not a magnitude: not a quantity that has diminished, but a discrimination the system can no longer make. What follows from that—and in particular whether a congruence can be made to evolve without circularity, so that a system might recover a distinction it had lost—is the subject of the next part.

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